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AIM

To perform statistical analysis and hypothesis testing on a real-world dataset using Python in order to identify relationships between variables, validate assumptions, and support data-driven decision-making.

STEP 1: IMPORT REQUIRED LIBRARIES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
import statsmodels.api as sm
from statsmodels.formula.api import ols

import tkinter as tk
from tkinter import filedialog

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

STEP 2: SELECT DATASET FROM COMPUTER

```
from google.colab import files
import io

print("Please upload the IBM HR Employee Attrition Dataset (CSV file).")
uploaded = files.upload()

if not uploaded:
    print("No file uploaded. Program stopped.")
    exit()

# Assuming only one file is uploaded
file_name = next(iter(uploaded))
file_path = io.BytesIO(uploaded[file_name])

print(f"\nSelected File: {file_name}")

# Load dataset
df = pd.read_csv(file_path)

print("\nDataset loaded successfully!")
print("Dataset Shape:", df.shape)
```

```
Please upload the IBM HR Employee Attrition Dataset (CSV file).
Choose Files No file chosen  Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving WA_Fn-UseC_-HR-Employee-Attrition.csv to WA_Fn-UseC_-HR-Employee-Attrition.csv

Selected File: WA_Fn-UseC_-HR-Employee-Attrition.csv

Dataset loaded successfully!
Dataset Shape: (1470, 35)
```

STEP 3: UNDERSTAND THE DATASET

```
print("\n========== FIRST 5 ROWS ==========")
print(df.head())

print("\n========== DATASET INFORMATION ==========")
print(df.info())

print("\n========== MISSING VALUES ==========")
print(df.isnull().sum())

print("\n========== DUPLICATE ROWS ==========")
print(df.duplicated().sum())
```

```
========== FIRST 5 ROWS ==========
   Age Attrition       BusinessTravel DailyRate              Department \
0   41       Yes      Travel_Rarely       1102                   Sales
1   49        No  Travel_Frequently        279 Research & Development
2   37       Yes      Travel_Rarely       1373 Research & Development
3   33        No  Travel_Frequently       1392 Research & Development
4   27        No      Travel_Rarely        591 Research & Development

     DistanceFromHome Education EducationField EmployeeCount EmployeeNumber \
0                   1         2  Life Sciences             1              1
1                   8         1  Life Sciences             1              2
2                   2         2          Other             1              4
3                   3         4  Life Sciences             1              5
4                   2         1        Medical             1              7

   ...  RelationshipSatisfaction StandardHours StockOptionLevel \
0 ...                         1            80                0
...   2                         4            80                1
...   3                         2            80                0
...   4                         3            80                0
...                             4            80                1

   TotalWorkingYears TrainingTimesLastYear WorkLifeBalance YearsAtCompany \
0                  8                     0               1              6
1                 10                     3               3             10
2                  7                     3               3              0
3                  8                     3               3              8
4                  6                     3               3              2

   YearsInCurrentRole YearsSinceLastPromotion YearsWithCurrManager
0                   4                       0                    5
1                   7                       1                    7
2                   0                       0                    0
3                   7                       3                    0
4                   2                       2                    2

[5 rows x 35 columns]

========== DATASET INFORMATION ==========
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
     Age                     0  1470   non-null int64
     Attrition               1  1470   non-null object
     BusinessTravel          2  1470   non-null object
     DailyRate               3  1470   non-null int64
     Department              4  1470   non-null object
     DistanceFromHome        5  1470   non-null int64
     Education               6  1470   non-null int64
     EducationField          7  1470   non-null int64
     EmployeeCount           8  1470   non-null object
     EmployeeNumber          9  1470   non-null int64
                                1470   non-null int64
                                1470   non-null int64
   10 EnvironmentSatisfaction  1470 non-null int64
                   11 Gender           non-null object
 12 HourlyRate                         int64
```

STEP 4: DESCRIPTIVE STATISTICS

```
# Select important numerical columns
numeric_columns = [
    "Age",
    "MonthlyIncome",
    "YearsAtCompany",
    "YearsInCurrentRole",
    "JobSatisfaction",
    "JobLevel"
]

# Keep only columns that actually exist in the dataset
numeric_columns = [col for col in numeric_columns if col in df.columns]

print("\n========== DESCRIPTIVE STATISTICS ==========")

for column in numeric_columns:

    print("\n----------------------------------------")
    print("Column:", column)

    print("Mean:",               df[column].mean())
    print("Median:",           df[column].median())
    print("Mode:",            df[column].mode()[0])
    print("Variance:",           df[column].var())
    print("Standard Deviation:", df[column].std())
```

```
========== DESCRIPTIVE STATISTICS ==========

----------------------------------------
Column: Age
Mean: 36.923809523809524
Median: 36.0
Mode: 35
Variance: 83.45504878602227
Standard Deviation: 9.135373489136734


----------------------------------------
Column: MonthlyIncome
Mean: 6502.931292517007
Median: 4919.0
Mode: 2342
Variance: 22164857.07151842
Standard Deviation: 4707.956783097995


----------------------------------------
Column: YearsAtCompany
Mean: 7.008163265306122 5
Median: 5.0
Mode: 5
Variance: 37.53431044303354
Standard Deviation: 6.126525152403566


----------------------------------------
Column: YearsInCurrentRole
Mean: 4.229251700680272
Median: 3.0
Mode: 2
Variance: 13.127121972001953
Standard Deviation: 3.6231370346706395


----------------------------------------
Column: JobSatisfaction
Mean: 2.7285714285714286
Median: 3.0
Mode: 4
Variance: 1.2162695711368152
Standard Deviation: 1.1028461230547149


----------------------------------------
Column: JobLevel
Mean: 2.0639455782312925
Median: 2.0
Mode: 1
Variance: 1.2253159398544744
Standard Deviation: 1.1069398989351114
```

STEP 5: PEARSON CORRELATION ANALYSIS

```
print("\n========== PEARSON CORRELATION MATRIX ==========")

correlation_matrix = df[numeric_columns].corr(method="pearson")

print(correlation_matrix)

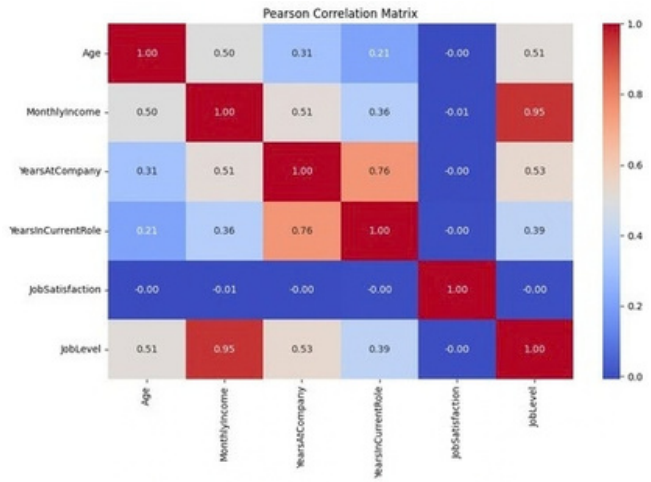
# Plot correlation heatmap
plt.figure(figsize=(10, 7))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Pearson Correlation Matrix")
plt.tight_layout()
plt.show()
```

```
========== PEARSON CORRELATION MATRIX ==========
                          Age MonthlyIncome YearsAtCompany \
Age                  1.000000       0.497855        0.311309
MonthlyIncome        0.497855       1.000000        0.514285
YearsAtCompany       0.311309       0.514285        1.000000
YearsInCurrentRole 0.212901         0.363818        0.758754
-J0o.b0S0a4t8i9s2faction           -0.007157       -0.003803
J0o.b5L0e9v6e014                    0.950300        0.534739

                     YearsInCurrentRole   JobSatisfaction JobLevel
Age                            0.212901         -0.004892 0.509604
MonthlyIncome                  0.363818         -0.007157 0.950300
YearsAtCompany                 0.758754         -0.003803 0.534739
YearsInCurrentRole             1.000000         -0.002305 0.389447
JobSatisfaction               -0.002305         1.000000 -0.001944
JobLevel                       0.389447         -0.001944 1.000000
```



STEP 7: INDEPENDENT SAMPLE T-TEST

```
print("\n========== INDEPENDENT SAMPLE T-TEST ==========")

# Separate employees based on Attrition
stayed = df[df["Attrition"] == "No"]["MonthlyIncome"]
left = df[df["Attrition"] == "Yes"]["MonthlyIncome"]

# Perform independent sample t-test
t_statistic, p_value = stats.ttest_ind(
    stayed,
    left,
    equal_var=False
)

print("Mean Monthly Income - Stayed :", stayed.mean())
print("Mean Monthly Income - Left   :", left.mean())

print("\nT-Statistic:", t_statistic)
print("P-Value:", p_value)

# Significance level
alpha = 0.05

print("\nSignificance Level (Alpha):", alpha)

if p_value < alpha:
    print("Result: Reject the Null Hypothesis (H0)")
    print("There is a statistically significant difference in Monthly Income.")
else:
    print("Result: Fail to Reject the Null Hypothesis (H0)")
    print("There is no statistically significant difference in Monthly Income.")
```

```
========== INDEPENDENT SAMPLE T-TEST ==========
Mean Monthly Income - Stayed : 6832.739659367397
Mean Monthly Income - Left   : 4787.092827004219 5

T-Statistic: 7.482621586644742
P-Value: 4.433588628286071e-13

Significance Level (Alpha): 0.05
Result: Reject the Null Hypothesis (H0)
There is a statistically significant difference in Monthly Income.
```

STEP 8: ONE-WAY ANOVA

```
print("\n========== ONE-WAY ANOVA ==========")

# Create groups of Job Satisfaction based on Job Role
groups = []

job_roles = df["JobRole"].dropna().unique()

for role in job_roles:

    satisfaction = df[
        df["JobRole"] == role
    ]["JobSatisfaction"].dropna()

    if len(satisfaction) > 1:
        groups.append(satisfaction)

# Perform One-Way ANOVA
anova_statistic, anova_p_value = stats.f_oneway(*groups)

print("Job Roles:", list(job_roles))

print("\nF-Statistic:", anova_statistic)
print("P-Value:", anova_p_value)

alpha = 0.05

print("\nSignificance Level (Alpha):", alpha)

if anova_p_value < alpha:
    print("Result: Reject the Null Hypothesis (H0)")
    print("Job Satisfaction differs significantly among Job Roles.")
else:
    print("Result: Fail to Reject the Null Hypothesis (H0)")
    print("No significant difference in Job Satisfaction among Job Roles.")
```

```
========== ONE-WAY ANOVA ==========
Job Roles: ['Sales Executive', 'Research Scientist', 'Laboratory Technician', 'Manufacturing Director', 'Healthcare Representative', 'Manager', 'Sales Representative', 'Research Director', 'Human Resources']

F-Statistic: 0.3652125640086401 6
P-Value: 0.9389893610579624

Significance Level (Alpha): 0.05
Result: Fail to Reject the Null Hypothesis (H0)
No significant difference in Job Satisfaction among Job Roles.
```

STEP 9: SIMPLE LINEAR REGRESSION

```
print("\n========== LINEAR REGRESSION ==========")

# Independent variable
X = df["YearsAtCompany"]

# Dependent variable
Y = df["MonthlyIncome"]

# Add constant/intercept
X_with_constant = sm.add_constant(X)

# Build regression model
model = sm.OLS(Y, X_with_constant).fit()

# Display regression summary
print(model.summary())

# Regression coefficients
print("\nIntercept:", model.params["const"])
print("Slope:", model.params["YearsAtCompany"])

# R-Squared
print("R-Squared:", model.rsquared)
```

```
========== LINEAR REGRESSION ==========
                            OLS Regression Results
==============================================================================
Dep. Variable:      MonthlyIncome   R-squared:                       0.264
Model:                        OLS   Adj. R-squared:  F-               0.264
Method:             Least Squares   statistic:                       527.9
Date:            Sun, 06 Sep 2026   Prob  (F-statistic):           4.82e-100
Time:                    14:39:39   Log-Likelihood: AIC:            -14291.
No. Observations:            1470   BIC:                            2.859e+04
Df Residuals:                1468                                   2.860e+04
Df Model:                       1
Covariance Type:        nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       3733.2731    160.091     23.320      0.000     3419.242     4047.305
YearsAtCompany   395.2046     17.201     22.976      0.000      361.464      428.945
==============================================================================
Omnibus:                      387.315   Durbin-Watson:                   2.034
Prob(Omnibus):                  0.000   Jarque-Bera  (JB):              875.316
Skew:                           1.462   Prob(JB):                     8.46e-191
Kurtosis:                       5.395   Cond. No.                         14.2
==============================================================================

Notes:
```

```
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Intercept: 3733.2731481323854
Slope: 395.20456923368227
R-Squared: 0.26448888197942455
```

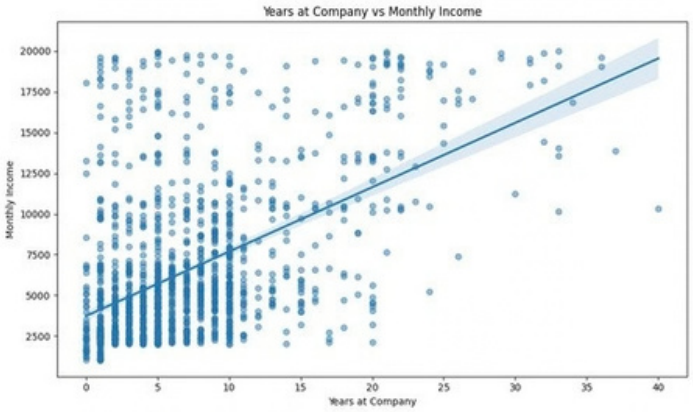
STEP 10: REGRESSION VISUALIZATION

```
plt.figure(figsize=(10, 6))

sns.regplot(
    x="YearsAtCompany",
    y="MonthlyIncome",
    data=df,
    scatter_kws={"alpha": 0.5}
)

plt.title("Years at Company vs Monthly Income")
plt.xlabel("Years at Company")
plt.ylabel("Monthly Income")

plt.tight_layout()
plt.show()
```



STEP 11: REGRESSION INTERPRETATION

```
print("\n========== REGRESSION INTERPRETATION ==========")

slope = model.params["YearsAtCompany"]
intercept = model.params["const"]
r_squared = model.rsquared

print("Regression Equation:")
print(
    f"MonthlyIncome = {intercept:.2f} + "
    f"({slope:.2f} × YearsAtCompany)"
)

print("\nR-Squared:", r_squared)

if slope > 0:
    print("Interpretation: Monthly Income tends to increase")
    print("as Years at Company increases.")
elif slope < 0:
    print("Interpretation: Monthly Income tends to decrease")
    print("as Years at Company increases.")
else:
    print("Interpretation: No linear relationship was found.")

print("\nConfidence Intervals:")
print(model.conf_int())
```

```
========== REGRESSION INTERPRETATION ==========
Regression Equation:
MonthlyIncome = 3733.27 + (395.20 × YearsAtCompany)

R-Squared: 0.26448888197942455
Interpretation: Monthly Income tends to increase
as Years at Company increases.

Confidence Intervals:
                        0           1
const          3419.241742  4047.304554
YearsAtCompany   361.463709   428.945429
```

STEP 12: FINAL SUMMARY

```
print("\n")
print("=" * 60)
print("               FINAL STATISTICAL SUMMARY")
print("=" * 60)

print("\n1. Dataset Shape:", df.shape)

print("\n2. T-TEST")
print("   T-Statistic:", t_statistic)
print("   P-Value:", p_value)

if p_value < 0.05:
    print("   Conclusion: Significant difference found.")
else:
    print("   Conclusion: No significant difference found.")

print("\n3. ANOVA")
print("   F-Statistic:", anova_statistic)
print("   P-Value:", anova_p_value)

if anova_p_value < 0.05:
    print("   Conclusion: Significant difference among groups.")
else:
    print("   Conclusion: No significant difference among groups.")

print("\n4. LINEAR REGRESSION")
print("   Intercept:", intercept)
print("   Slope:", slope)
print("   R-Squared:", r_squared)

print("\n5. Correlation Matrix:")
print(correlation_matrix)

print("\n" + "=" * 60)
print("Experiment 3 Completed Successfully!")
print("=" * 60)
```

```
============================================================
               FINAL STATISTICAL SUMMARY
============================================================

1. Dataset Shape: (1470, 35)
```

```
2. T-TEST
   T-Statistic: 7.482621586644742
   P-Value:             4.433588628286071e-13
   Conclusion: Significant difference found.

3. ANOVA
   F-Statistic: 0.365212564608640 16
   P-Value: 0.9380893610579624
   Conclusion: No significant difference among groups.

4. LINEAR REGRESSION
   Intercept: 3733.2731481323854
   Slope: 395.2045692336822
   R-Squared: 0.264488888197942455


5. Correlation Matrix:
                          Age MonthlyIncome YearsAtCompany \
Age                  1.000000       0.497855        0.311309
MonthlyIncome        0.497855       1.000000        0.514285
YearsAtCompany       0.311309       0.514285        1.000000
YearsInCurrentRole   0.212901       0.363818        0.758754
JobSatisfaction     -0.004892      -0.007157       -0.003803
JobLevel             0.509604       0.950300        0.534739

                     YearsInCurrentRole JobSatisfaction JobLevel
Age                            0.212901        -0.004892 0.509604
MonthlyIncome                  0.363818        -0.007157 0.950300
YearsAtCompany                 0.758754        -0.003803 0.534739
YearsInCurrentRole             1.000000        -0.002305 0.389447
JobSatisfaction               -0.002305         1.000000 -0.001944
JobLevel                       0.389447        -0.001944 1.000000

================================================================
Experiment 3 Completed Successfully!
================================================================
```

Questions and Answers

QUESTIONS AND ANSWERS

Q1. What is the difference between descriptive statistics and inferential statistics?

Answer:

Descriptive statistics summarize and describe the data using measures such as mean, median, mode, variance and standard deviation.

Inferential statistics use sample data to make conclusions or predictions about a larger population.

Example:

Mean salary is descriptive statistics.

Testing whether two salary groups are significantly different is inferential statistics.

Q2. Explain the concepts of the Null Hypothesis (H0) and Alternative Hypothesis (H1).

Answer:

The Null Hypothesis (H0) states that there is no significant difference or relationship between variables.

The Alternative Hypothesis (H1) states that there is a significant difference or relationship.

Example:

H0: There is no significant difference in Monthly Income between employees who stayed and employees who left.

H1: There is a significant difference in Monthly Income between employees who stayed and employees who left.

Q3. What is a p-value? How is it used to make statistical decisions?

Answer:

A p-value indicates how likely it is to observe the obtained result if the Null Hypothesis is true.

Usually, the significance level is 0.05.

If p-value < 0.05:

Reject H0.

If p-value >= 0.05:

Fail to reject H0.

Therefore, a smaller p-value provides stronger evidence against the Null Hypothesis.

Q4. Differentiate between a t-test and ANOVA. In which situations is each test applied?

Answer:

A t-test is generally used to compare the means of two groups.

Example:

Comparing Monthly Income of employees who stayed and left.

ANOVA is used to compare means among three or more groups.

Example:

Comparing Job Satisfaction across different Job Roles.

Therefore:

t-test -> Two groups

ANOVA -> Three or more groups

Q5. What does the Pearson correlation coefficient indicate? What are its possible values?

Answer:

Pearson correlation coefficient measures the strength and direction of a linear relationship between two numerical variables.

Its values range from -1 to +1.

+1 -> Perfect positive correlation

0 -> No linear correlation

-1 -> Perfect negative correlation

A positive value means both variables generally increase together.

A negative value means one variable generally increases while the other decreases.

Q6. Explain the significance of R^2 (Coefficient of Determination) in Linear Regression.

Answer:

R^2 represents the proportion of variation in the dependent variable that is explained by the independent variable.

R^2 ranges from 0 to 1.

For example:

$R^2 = 0.70$ means approximately 70% of the variation in the dependent variable is explained by the regression model.

A higher R^2 generally indicates a better fit to the data, although a high R^2 alone does not prove that the model is appropriate.

Q7. Why is statistical analysis important before applying machine learning algorithms?

Answer:

Statistical analysis helps understand the dataset before building a machine learning model.

It canR ideelanttioifny:ships between variables

- Outliers
- Missing values
- Data distribution
- Important features
- Statistical significance

This helps in selecting suitable features and algorithms and can improve the quality of machine learning models.

Q8. What assumptions should be satisfied before performing a t-test or ANOVA?

Answer:

Important assumptions include:

1. Observations should be independent.
2. Data should be approximately normally distributed within the groups.
3. For the standard versions of these tests, variances should be reasonably similar between groups.
4. The dependent variable should generally be numerical.

If these assumptions are strongly violated, alternative statistical tests may be considered.

Q9. How can regression analysis help organizations in forecasting and decision-making?

Answer:

Regression analysis helps organizations understand relationships between variables and make predictions.

For example:

An organization can analyze the relationship between employee experience and income.

Regression can also be used for:

- Sales forecasting
- Revenue prediction
- Demand forecasting
- Cost estimation

- Employee analytics

Therefore, regression supports data-driven decision-making.

Q10. Give two real-world applications where hypothesis testing is commonly used in data analytics.

Answer:

1. Business:
   A company can test whether a new marketing campaign significantly increases sales compared with the old campaign.
2. Healthcare:
   Researchers can test whether a new treatment produces significantly different results compared with an existing treatment.

Other applications include finance, manufacturing, education, customer analytics and quality control.

END OF EXPERIMENT 3